

Cyber Security Task 02: CIA Triad & Risk Assessment

Objective

The purpose of this task is to understand the core principles of Cyber Security and learn how organizations identify and manage security risks.

The CIA Triad is the foundation of all cybersecurity practices.

Part A: Understanding the CIA Triad

Research and explain:

Confidentiality

- Definition
- Why it is important
- Real-world example

Integrity

- Definition
- Why it is important
- Real-world example

Availability

- Definition
- Why it is important
- Real-world example

Create a comparison table for all three concepts.

Part B: Real-World Case Study

Choose any one cyber attack from the following:

- WannaCry Ransomware
- Equifax Data Breach
- SolarWinds Attack
- Colonial Pipeline Attack
- Facebook Data Leak

Explain:

1. What happened?
2. Which part of the CIA Triad was affected?
3. What was the impact?
4. How could it have been prevented?

Write in your own words.

Part C: Risk Assessment Activity

Imagine you are a Security Analyst for a college.

Identify at least 5 assets that need protection.

Example:

- Student Records
- Examination Data
- College Website
- Faculty Information
- Network Infrastructure

For each asset identify:

Asset	Threat	Impact	Risk Level
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Example:

| Student Records | Data Breach | High | High |

Create a complete risk assessment table.

Part D: Threat Identification

Research and explain the following threats:

1. Insider Threat
2. Malware
3. Phishing
4. Weak Passwords
5. Unpatched Systems

For each threat include:

- Description

- Possible Impact
 - Prevention Methods
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Part E: Security Recommendations

Based on your risk assessment, suggest at least:

- 10 Security Recommendations

Examples:

- Enable Multi-Factor Authentication
- Regular System Updates
- Strong Password Policies
- Employee Security Training
- Regular Backups

Explain why each recommendation is important.

Part F: Mini Security Consultant Activity

Imagine your college asks you:

"How can we improve our cybersecurity posture?"

Prepare a one-page report containing:

- Top 5 Risks
- Top 5 Recommendations
- Expected Benefits

Write professionally.

Documentation

Create a folder named:

`text id="nj7yzj"`
`CyberSecurity_Task_02_YourName`

Inside the folder include:

- CIA Triad Research
- Cyber Attack Case Study
- Risk Assessment Table

- Threat Analysis
- Security Recommendations
- Consultant Report
- README.md

Upload everything to GitHub.

Submission

The purpose of this task is learning and practical thinking.

This is not a marks-based task.

Submit your updated GitHub repository link through the Sunday Submission Form before the weekly deadline.

Continue maintaining all internship tasks in the same repository using separate folders.

Topics to Study Before the Next Session

Please spend some time learning:

- CIA Triad
- Risk Management
- Vulnerability vs Threat vs Risk
- Security Controls
- Types of Malware
- Data Protection
- Security Policies
- Cybersecurity Framework Basics
- Incident Response Fundamentals

Recommended Study Time:

30–45 minutes per day.

Remember: Before learning advanced security tools and ethical hacking techniques, every cybersecurity professional must understand what they are protecting, why it needs protection, and what risks exist.